



THE ROMAN CATHOLIC DIOCESE OF SUMBAWANGA

**FORM FOUR MOCK EXAMINATION FOR  
DIOCESAN SECONDARY SCHOOLS 2026**



032/2A

CHEMISTRY 2A

Time 2:30 Hours

2026

**Instructions**

1. This paper consists of two (2) questions. Answer all questions
2. Each question carries twenty five (25) marks
3. Qualitative analysis guide sheet authorized by NECTA and non-programmable calculators may be used.
4. Cellular phones and any unauthorized materials are not allowed in the examination room
5. Write your Examination number on every page of your answer sheets
6. The following constant may be used.

Na = 23, H = 1, S = 32, O = 16, Cl = 35.5, C = 12, N = 14

Atomic masses, H=1, Cl=35.5, Na= 23, C = 12, O = 16.

1 dm<sup>3</sup> = 1 litre = 1000 cm<sup>3</sup>

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1. You are provided with:

AA: 3.65g of Hydrochloric acid, HCl in 1 litre of solution.

BB: 1.20 g of impure sodium carbonate, anhydrous,  $\text{Na}_2\text{CO}_3$  in 250cm<sup>3</sup> of solution.

Methyl orange indicator

**PROCEDURE:**

- Pipette 20cm<sup>3</sup> (or 25 cm<sup>3</sup>) of solution BB into a clean conical flask. Add 3 drops of methyl orange indicator, and titrate with solution AA from the burette until the solution is just pink and record the reading in the trial column (pilot) of your table of results.
- Repeat the titration with other three portions of the solution BB to three successive burette readings.

**QUESTIONS:**

- Calculate the average volume of solution AA used.
- ..... cm<sup>3</sup> of AA required.....cm<sup>3</sup> of BB for complete reaction.
- Write a well-balanced chemical reaction between solutions AA and BB.
- Write the ionic equation of the reaction between AA and BB.
- Calculate the concentration of solution AA in mol dm<sup>-3</sup>.
- Calculate the concentration of solution BB in :
  - mol dm<sup>-3</sup>
  - g dm<sup>-3</sup>
- Calculate the percentage purity of sodium carbonate, (BB) in the solution.

2. You are provided with the following:

DD: A solution containing 20.54g/dm<sup>3</sup> of anhydrous  $\text{Na}_2\text{S}_2\text{O}_3$

LL: A solution contain 2 M HCl.

Distilled water

Stop watch/clock

**Procedure**

- Mark letter X on a piece of paper
- Measure 50 cm<sup>3</sup> of DD and pour it into a conical flask
- Measure 10cm<sup>3</sup> of LL by using different measuring cylinder pour in conical flask containing DD immediately start the stop watch and observe the changes from above
- Stand the conical flask on the paper above the cross. Look down vertically through the mouth of the conical flask and make sure you can see the cross at the bottom of the flask.
- Record the time taken for the disappearance of letter X
- Repeat the whole procedure using 40 cm<sup>3</sup>, 30 cm<sup>3</sup>, 20 cm<sup>3</sup> and finally 10 cm<sup>3</sup>.

Table 1.

| Number of experiment | Volume of D (cm <sup>3</sup> ) | Volume of distilled H <sub>2</sub> O (cm <sup>3</sup> ) | Concentration of Na <sub>2</sub> S <sub>2</sub> O <sub>3</sub> (mol/dm <sup>3</sup> ) | Time (t) in second | Rate of reaction(S <sup>-1</sup> ) |
|----------------------|--------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------|------------------------------------|
| 1                    | 50                             | 0                                                       |                                                                                       |                    |                                    |
| 2                    | 40                             | 10                                                      |                                                                                       |                    |                                    |
| 3                    | 30                             | 20                                                      |                                                                                       |                    |                                    |
| 4                    | 20                             | 30                                                      |                                                                                       |                    |                                    |
| 5                    | 10                             | 40                                                      |                                                                                       |                    |                                    |

### QUESTIONS

- Complete by filling the Table 1
- Write the balance chemical equation between DD and LL.
- What substance was produced during the reaction which obscured the mark X?
- What conclusion can you draw from the result of this experiments?
- Briefly explain the smell and color observed as the reaction take place.
- Explain how concentration changes affect the rate of chemical reaction.
- Plot the graph of time (y-axis) against Concentration of Na<sub>2</sub>S<sub>2</sub>O<sub>3</sub> (x- axis)